Name:

UCInetID:

Data 275P: Fall 2026

Machine Learning with Generative Models

Homework 2

Due Date: **Monday, October 26th**

As with previous Homeworks, this homework includes a mix of code that should be submitted to the GradeScope autograder, and a templated PDF that should be filled out with mathematical derivations, plots, and numerical answers. Please see Canvas for more details, but as always please take care that:

- For the PDF submission, you must **fill in this PDF template** to provide your answers. All pages should remain in their original order when uploading your assignment to gradescope. *Do not rearrange, add, or remove any pages from the provided PDF template.*
- Enter your final answers in the dedicated answer spaces, and write your explanations and/or any math equations in the space provided below each question. *Always show the work needed to produce your answers, not just the final result.* Although typed responses are preferred, legible handwritten responses are fine.
- For multiple-choice questions, be sure to clearly select one of the answers, as well as providing an explanation or justification to support your answer.
- Check that all answers are readable, all scanned math or imported images are clear, and that content has not been cut off by the page margins. (We can only grade answers that we can see!)
- A Jupyter notebook version of the homework is also provided with each assignment.
- Some responses require inclusion of a plot or figure; after generating the plot in python, you can export it to a file or copy it to your clipboard, and then paste it into the homework PDF where indicated.
- If you would like to include code snippets in your responses, you can do the same thing – take an image of your code, and import it into the PDF in the space provided, or paste it into a textbox in the PDF.

If you have any questions/concerns about the homework problems or using Jupyter notebooks, ask us on EdD.

Summary of Assignment: 100 total points

- Problem 1: Parameter Estimation (35 points)
- Problem 2: Baysian Logistic Regression (40 + 25 points)

Problem 1: Parameter Estimation (35 points)

This question asks you to devise ML and Bayesian MAP estimators for a simple model of an uncalibrated sensor. Let the sensor output, X, be a random variable that ranges over the positive real numbers. We assume that, when tested over a range of environments, its outputs are uniformly distributed on some unknown interval $[0, \theta]$, so that

$$p(x \mid \theta) = \frac{1}{\theta} \mathbb{1}[0 \le x \le \theta] = \begin{cases} 1/\theta & \text{if } 0 \le x \le \theta, \\ 0 & \text{otherwise.} \end{cases}$$

Here, $\mathbb{1}[0 \le x \le \theta]$ is an *indicator function* that equals 1 when the condition is true, and 0 otherwise. We denote this distribution by $\text{Unif}(0, \theta)$. To calibrate the sensor, we would like to infer θ.

1. Given N i.i.d. observations $x = (x_1, \ldots, x_N)$, $X_i \sim \text{Unif}(0, \theta)$, what is the likelihood function $p(x \mid \theta)$? What is the maximum likelihood (ML) estimator for θ? Give an explanation for why your estimator is in fact the ML estimator.

Likelihood: $p(x \mid \theta) =$

ML estimate: $\hat{\theta} =$

Brief explanation for how the ML estimate is derived:

2. Suppose that we place the following prior distribution on θ:

$$p(\theta) = \alpha\beta^{\alpha}\theta^{-\alpha-1}\,\mathbb{1}[\beta \le \theta \le \infty].$$

This is known as a *Pareto distribution*. We denote it by $\theta \sim \text{Pareto}(\alpha, \beta)$. Plot the three prior probability densities corresponding to the following three hyperparameter choices: $(\alpha, \beta) = (0.1, 0.1)$; $(\alpha, \beta) = (2.0, 0.1)$; $(\alpha, \beta) = (1.0, 2.0)$. Briefly describe the influence these parameters have on the properties of the Pareto distribution.

Plot, Pareto(α, β):

Role of α, β in the distribution?

3. If $\theta \sim \text{Pareto}(\alpha, \beta)$ and we observe N uniformly distributed observations $X_i \sim \text{Unif}(0, \theta)$, derive the posterior distribution $p(\theta \mid x)$. Is this a member of any standard family?

Posterior, $p(x|\theta) =$

Derivation of posterior distribution, including alignment with standard family:

4. For the posterior derived in part (c), what is the corresponding MAP estimator of θ? How does this compare to the ML estimator?

MAP estimate: $\hat{\theta} =$

Brief discussion of the relationship between the MAP and ML parameter estimates:

5. Recall that the quadratic loss is defined as $L(\theta, \hat{\theta}) = (\theta - \hat{\theta})^2$. For the posterior derived in part (c), what estimator of θ minimizes the posterior expected quadratic loss? Simplify your answer as much as possible.

MMSE estimate: $\hat{\theta} =$

Derivation of the minimum mean squared error (MMSE) parameter estimate:

Aside: *A recent aphorism coined by Nathan Calvin after a few high-profile AI safety failures was, "If you discover two ants in your kitchen, the best estimate of the total number of ants in your kitchen is not two."*

6. Suppose that we observe three observations $x = (0.7, 1.3, 1.7)$. Determine the posterior distribution of θ for each of the priors in part (b), and plot the corresponding posterior densities. What is the MAP estimate for each hyperparameter choice? What estimate minimizes the quadratic loss for each hyperparameter choice?

Plot, posterior densities for three priors:

Parameter estimates minimizing expected loss for each choice of prior distribution:

$(\alpha, \beta) = (0.1, 0.1) : \hat{\theta}_{\text{MAP}} =$

$(\alpha, \beta) = (0.1, 0.1) : \hat{\theta}_{\text{MMSE}} =$

$(\alpha, \beta) = (2.0, 0.1) : \hat{\theta}_{\text{MAP}} =$

$(\alpha, \beta) = (2.0, 0.1) : \hat{\theta}_{\text{MMSE}} =$

$(\alpha, \beta) = (1.0, 2.0) : \hat{\theta}_{\text{MAP}} =$

$(\alpha, \beta) = (1.0, 2.0) : \hat{\theta}_{\text{MMSE}} =$

Problem 2: Bayesian Logistic Regression (40 + 25 points)

In this question, we use multinomial logistic regression models to predict one of K discrete classes. Let $t_{nk} = 1$ if observation n is an instance of class k, and $t_{nk} = 0$ otherwise. Given input variable x_n, let $\phi(x_n) \in \mathbb{R}^M$ be a fixed feature function. The multinomial logistic regression model then defines the conditional probability of class k:

$$p(t_{nk} = 1 \mid x_n, w) = \frac{\exp(w_k^T \phi(x_n))}{\sum_{\ell=1}^{K} \exp(w_\ell^T \phi(x_n))},$$

where $w_k \in \mathbb{R}^M$ are the feature weights of class k, and $w \in \mathbb{R}^{KM}$ is a vector concatenating the weights for all classes.

This model is not identifiable: matching translations of the weight vectors w_k for different classes can produce identical probabilities. To avoid this ambiguity and improve robustness, we place a zero-mean, diagonal-covariance Gaussian prior on the weight vector:

$$p(w) = \mathcal{N}(w \mid 0, \alpha^{-1} I_{KM}) \propto \exp\left(-\frac{\alpha}{2} w^T w\right).$$

Here, α is a tunable inverse-variance parameter that controls the degree of regularization. The negative log-posterior distributiona (whose minimum corresponds to the MAP estimate of the weight vector w) is then

$$\begin{aligned} f(w) &= C - \log p(w) - \sum_{n=1}^{N} \log p(t_n \mid x_n, w) \\ &= \frac{\alpha}{2} w^T w - \sum_{n=1}^{N} \left[\sum_{k=1}^{K} t_{nk} w_k^T \phi(x_n) - \log\left(\sum_{\ell=1}^{K} \exp(w_\ell^T \phi(x_n)) \right) \right], \end{aligned}$$

ignoring constants C which do not depend on w. Taking derivatives with respect to some weight w_{km} and simplifying, we can then show that

$$\frac{\partial f(w)}{\partial w_{km}} = \alpha w_{km} - \sum_{n=1}^{N} (t_{nk} - \mu_{nk}) \phi_m(x_n), \qquad \mu_{nk} = \frac{\exp(w_k^T \phi(x_n))}{\sum_{\ell=1}^{K} \exp(w_\ell^T \phi(x_n))}.$$

These derivatives define the gradients needed for MAP estimation via gradient descent.

We reexamine the gamma ray data set from Homework 1, but instead apply a logistic regression model for the binary classification of star showers. We use the same split of the data into 15,216 training examples and 3,804 test examples. We compare the performance of three different feature mappings of the $D = 10$ raw inputs. In all cases, the first feature should be a constant bias or offset term, $\phi_0(x_n) = 1$. The three feature sets are then:

1. $D + 1$ linear features, the bias feature plus the raw inputs $\phi_m(x_n) = x_{nm}, 1 \le m \le D$.
2. $2D + 1$ diagonal quadratic features, including the $D + 1$ linear features from set 1, as well as D quadratic features $\phi_{D+m}(x_n) = x_{nm}^2, 1 \le m \le D$.
3. $(D + 1)D/2 + D + 1$ general quadratic features, including the $2D + 1$ features from set 2, as well as products of all pairs of input dimensions $x_{nm} x_{nm'}, m \ne m'$.

Given this data and the objective from part (a), we will use the `minimize` function from the `scipy.optimize` package to find the weight vector w that minimizes the negative log-posterior distribution $f(w)$. You should provide the optimizer with the gradient of $f(w)$, and write your *own* function to compute the objective $f(w)$ and its gradient $\nabla f(w)$. (For this part, you may *not* simply use numerical approximations to the gradient.) To get started with `minimize`, see the sample code released with the homework, which includes suggestions for appropriate convergence tolerance parameters.

For all of the following questions, set the regularization constant to $\alpha = 10^{-6}$, and test each of the three feature sets defined above.

1. For each of the three feature sets, train the logistic regression model by running gradient-based optimization to convergence. Report the accuracy and negative log-posterior probability of the classifier when training and evaluating on `train`.

Error Rate, Linear Features:	Negative Log-Posterior, Linear Features:
Error Rate, Diagonal Quadratic:	**Negative Log-Posterior, Diagonal Quadratic:**
Error Rate, All Quadratic:	**Negative Log-Posterior, All Quadratic:**

2. For each of the three feature sets, and the MAP weight estimates from part (a), evaluate and report the `test` accuracy and `test` negative log-posterior probability.

Error Rate, Linear Features:	Negative Log-Posterior, Linear Features:
Error Rate, Diagonal Quadratic:	**Negative Log-Posterior, Diagonal Quadratic:**
Error Rate, All Quadratic:	**Negative Log-Posterior, All Quadratic:**

3. Compare the test accuracies of the logistic regression models to the Gaussian naive Bayes classifier from Homework 1. Which method is more accurate? Using concepts from lecture, briefly discuss possible reasons for the observed performance differences.

4. Using the provided homework code (based on the logistic regression Jupyter demo), run MCMC to get a collection of samples from the posterior distribution over weights. Use these samples to estimate the posterior (marginal) probability of each data point. Then, report the error (misclassification) rate of these predictions on the training and test data by thresholding them at probability 0.5. (Note that you do not need to compute the log likelihood for this part.)

Training Error Rate, Linear:

Training Error Rate, Diagonal Quadratic:

Training Error Rate, All Quadratic:

Testing Error Rate, Linear:

Testing Error Rate, Diagonal Quadratic:

Testing Error Rate, All Quadratic: